SRUJANA 2026 · PROMPT TO PRODUCT

Campus Chakra

Project Overview · From campus resource complexity to an evaluator-ready prototype

Initial idea

Campuses need a single operational view of electricity, water and waste, but real facilities data is noisy, delayed and distributed across sensor feeds and manual notes. Campus Chakra was conceived as an edge-native circular-resource grid that turns these disconnected signals into timely, understandable actions.

Development process

1. Read the competition problem statement and all provided data files.
2. Mapped data constraints to four workflows: ingestion integrity, waste classification, short-horizon demand outlook and facility action.
3. Implemented backend endpoints with explicit upload validation and deterministic fallback behavior.
4. Built the Campus Chakra dashboard experience, guide flow and brand entrance.
5. Ran local smoke tests with the supplied competition files and produced the required test predictions.

Implemented features

Reliable data intake

Checks schemas, timestamps, numeric telemetry and known facilities before analysis.

Smart-bin inference

Predicts a waste stream from sensor fields and exports competition-ready predictions.

Forecast and alerts

Transforms past demand into 6h / 12h planning signals and facility follow-ups.

Tools used

Layer	Tools
Backend and data	Python, FastAPI, pandas, scikit-learn
Frontend	HTML, CSS, JavaScript, Chart.js, GSAP, Lenis
AI-assisted development	Codex CLI and product-integrated LLM providers with deterministic fallbacks
Delivery validation	Local endpoint checks, CSV verification, PDF generation and UI capture

Final outcome

> **Campus Chakra is a working competition prototype:** jurors can upload the supplied-format data, receive classification and forecast outputs, inspect facility workflows and review a clean submission package. The design intentionally keeps essential intelligence available locally so the demo remains functional even when optional external AI services are unavailable.

What makes it useful

- Combines resource telemetry and waste operations instead of treating them as isolated dashboards.
- Prioritizes explainable action: each alert has a relevant evidence path and next step.
- Maintains data integrity through validation and idempotent synchronization.
- Creates a direct bridge from sensor classification to circular-resource allocation.